

AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic

Complete Technical Brief, Forensic Deep Dive & End-to-End Offline Architecture Blueprint

PROBLEM STATEMENT CODE
SIH26146 (#146 of 236)

SPONSORING AGENCY
NTRO (PMO / NSA)

THEME & DOMAIN
Blockchain &
Cybersecurity

AWARD & PLATFORM
₹ 1,00,000 / Linux
(Offline)

1. Executive Summary & Problem Context

Problem Statement SIH26146 is sponsored by the **National Technical Research Organisation (NTRO)**, India's premier technical intelligence organisation responsible for cyber intelligence, national cryptographic security, and forensic surveillance under the National Security Advisor (NSA) in the Prime Minister's Office.

The Core Threat: While Bitcoin provides an immutable, transparent public distributed ledger, its pseudonymous and peer-to-peer (P2P) architecture makes it the favored vehicle for cybercrime cartels, ransomware extortion syndicates, darknet narcotics vendors, and terror-financing networks. Malicious actors employ sophisticated obfuscation typologies—such as *CoinJoin mixing*, *recursive peeling chains*, *multi-hop burner wallets*, and *proxy/Tor tunnels*—to decouple real-world identities from transaction flows.

The Intelligence Blindspot: Traditional signals intelligence monitors *Network-Layer observations* (IP addresses, TCP ports, handshake latencies, Autonomous System Numbers), whereas traditional blockchain forensics only inspects *Blockchain-Layer records* (UTXOs, public wallet addresses, and transaction hashes). There is currently an operational void: **how to bridge the gap between network-layer metadata and on-chain ledger activity without decrypting encrypted payloads?**

Official NTRO Challenge Objective

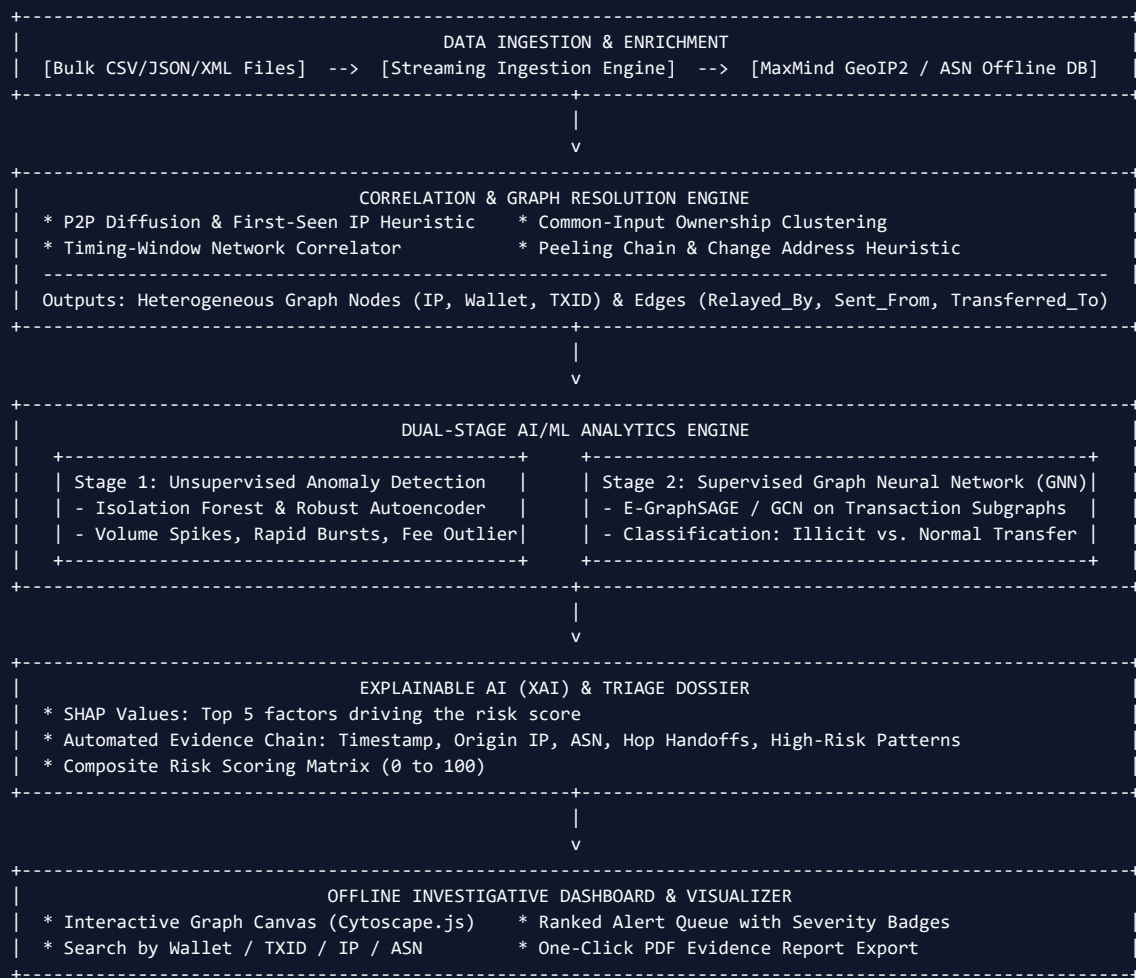
Design and build a **complete, workable offline system for Linux** that ingests bulk Bitcoin transaction and network metadata (CSV, JSON, XML), correlates network-layer (IP/port/timing) observations with blockchain-layer (wallet/TXID/amount) data, and applies **real AI/ML models** (not just static if-else rules) to detect anomalies, cluster entities, and generate prioritized, explainable investigative leads with confidence scores.

2. Deep Problem Analysis: Key Architectural & Forensic Challenges

CHALLENGE AREA	TECHNICAL REALITY & COMPLEXITY	FORENSIC / HACKATHON IMPACT
1. Network-Ledger Correlation	Bitcoin nodes propagate transactions across a peer-to-peer gossip network using diffusion timers. Multiple nodes broadcast the same TXID at slightly different times. Pinpointing the actual entry node requires statistical diffusion tree reconstruction.	Investigators need reliable, probabilistic attribution linking a transaction hash to an originating IP address and physical Autonomous System (ASN).
2. Laundering Typologies & Evasion	Adversaries utilize peeling chains (breaking large funds into 100+ micro-hops) and CoinJoin/Whirlpool mixes (combining multiple inputs into identical outputs) to destroy graph linearity.	Simple rule-based thresholds fail and generate excessive false positives. System requires Graph Neural Networks (GNNs) capable of topological sub-graph pattern learning.
3. 100% Offline Linux Operation	NTRO mandates a <i>fully offline Linux solution</i> . No external API queries to Block Explorers, Chainalysis, or cloud endpoints are permitted.	All data parsing, entity clustering, GeoIP/ASN resolution, and ML inference must execute purely on local system memory and offline databases.
4. Court-Admissible Explainability (XAI)	A black-box neural network providing "Risk Score: 0.95" cannot be submitted as evidence for a legal seizure, judicial warrant, or inter-agency report.	The solution must provide transparent, feature-attributable reasons (via SHAP values, flow path graphs, and heuristic evidence flags).
5. High-Throughput Bulk Ingestion	Metadata arrives in massive, heterogeneous batches (CSV/JSON/XML) containing hundreds of thousands of transactions.	Requires high-speed streaming columnar parsers (Polars/DuckDB) to prevent memory crashes on standard workstation hardware.

3. End-to-End System Architecture Blueprint

The proposed architecture follows an offline, multi-tier intelligence pipeline designed to satisfy all NTRO constraints:



4. Technical Deep-Dive: Core Modules

Module 1: High-Performance Offline Ingestion & Network Enrichment

- **Streaming Multi-Format Ingestion:** Utilizes Polars (Rust-backed columnar engine) or DuckDB for lightning-fast parsing of CSV, JSON, and XML metadata files without memory overflow.
- **Offline GeoIP & ASN Resolution:** Integrates local, offline MaxMind GeoLite2 City and ASN databases (.mmdb format). Every source/destination IP is enriched with ISO country codes, Autonomous System Numbers (ASN), and ISP identities without requiring an internet connection.
- **Dataset Schema Handled:** timestamp, src_ip, dst_ip, src_port, dst_port, txid, input_addresses[], output_addresses[], input_amounts[], output_amounts[], fee, script_type (P2PKH, P2SH, SegWit, Taproot).

Module 2: Network-to-Blockchain Graph Correlation Engine

- **First-Seen Relay Heuristic:** Maps transaction broadcasts across peer nodes. The node with the earliest timestamp for a given TXID is identified as the likely *Entry-Node / Origin Node*.
- **Common-Input Ownership Clustering:** If a single transaction consumes inputs from address A, B, and C, all three addresses are mathematically clustered as belonging to the same entity/wallet owner.
- **Change Address Detection:** Distinguishes payment outputs from change returns using address format re-use, round-number payment heuristics, and script compatibility.
- **Tripartite Graph Structure:** Creates an in-memory directed multi-graph: [IP Node] --(BROADCASTS)--> [TXID Node] --(SPENDS FROM)--> [Wallet Address] --(OUTPUTS TO)--> [Destination Address].

Module 3: Dual-Stage AI/ML Detection Engine

NTRO explicitly demands a **working AI/ML model—not just static heuristic rules**. The solution utilizes a hybrid approach:

Component	Algorithm / Model	Target Illicit Pattern
Unsupervised Anomaly Detection	Isolation Forest & Deep Autoencoders	Detects statistical outliers: extreme transaction velocity, anomalous fee-to-volume ratios, irregular batch sizes, and sudden transaction bursts from previously dormant IPs.
Graph Topology Classification	Graph Neural Networks (GraphSAGE / GCN) & XGBoost	Trained on graph structural embeddings (node degree, centrality, clustering coefficient, in/out ratio, peeling depth) to identify laundering chains, mixer outputs, and ransomware cash-out clusters.
Temporal Sequence Analysis	Temporal Graph Attention Networks / LSTM	Captures time-decay patterns in peeling chains where funds are rapidly hopped through intermediary burner wallets within seconds.

Module 4: Explainable AI (XAI) & Investigative Lead Generation

To ensure investigative utility for intelligence officers, every alert is accompanied by an automated, court-ready evidentiary breakdown:

- SHAP (SHapley Additive exPlanations):** Computes exact quantitative contribution of each feature to the risk score (e.g., +35% due to 6-hop peeling chain topology, +25% due to high-risk ASN origin, +18% due to anomalous fee ratio).
- Composite Risk Scoring (0–100):** Risk Score = 0.40 * GNN_Score + 0.30 * Anomaly_Score + 0.30 * Heuristic_Risk_Factors.
- Automated Dossier Generation:** Generates structured investigative lead summaries in JSON and downloadable PDF formats containing full transaction lineage, associated IP geolocations, and recommended actions.

Module 5: Offline SOC-Style Web Dashboard & Visualization

- Interactive Network Canvas:** Built with **Cytoscape.js** or **Force-Directed Graphs**, enabling analysts to expand wallet nodes, trace fund hops, and visually inspect IP-to-wallet bindings.
- Triage Alert Feed:** Prioritized queue sorting entities into Medium Risk, High Risk, and Critical Malicious.
- Forensic Filters:** Filter by Country, ASN, Date-Range, Minimum BTC Value, Script Type, and Laundering Typology.

5. Suggested Technology Stack (Linux Offline Ready)

Component Layer	Recommended Technologies	Role & Justification
Core Language & Runtime	Python 3.11+, Bash Scripts, Docker (Linux Ubuntu/Debian)	Standard forensic runtime; native offline Docker packaging.
High-Speed Ingestion	Polars / DuckDB / PyArrow	Columnar processing capable of parsing 1M+ rows in seconds offline.
Offline Geo & Enrichment	GeoIP2 Python SDK, MaxMind GeoLite2-City & ASN (.mmdb)	100% offline IP-to-country and ASN mapping without API limits.
Graph Processing	NetworkX / Rustworkx / Neo4j Community (Offline)	Entity clustering, multi-input linking, shortest path tracing.
Machine Learning & XAI	PyTorch Geometric (PyG), Scikit-Learn, LightGBM, SHAP	Graph convolutional embeddings, anomaly isolation, explainability.
Backend API	FastAPI / Uvicorn (Asynchronous, lightweight)	High-concurrency offline RESTful endpoints for graph & leads.
Frontend UI & Visuals	React, Vite, TailwindCSS, Cytoscape.js / Vis-Network	Sleek, responsive dark-mode cyber intelligence dashboard.

6. Step-by-Step Implementation Roadmap for Hackathon

Phase 1: Ingestion & Enrichment Pipeline (Day 1 - Morning)

Set up local directory structure and download MaxMind offline databases. Implement `data_loader.py` using Polars to ingest sample synthetic CSV/JSON/XML data. Enrich every row with country code, ASN name, and normalized Bitcoin fields (Satoshis converted to BTC).

Phase 2: Graph Resolution & Entity Clustering (Day 1 - Afternoon)

Build `graph_builder.py`. Implement common-input heuristics to group multi-input addresses into unified clusters. Construct the heterogeneous network linking IP address nodes to TXID nodes and Wallet nodes. Calculate graph metrics (Degree, Betweenness Centrality, In-Degree/Out-Degree ratio).

Phase 3: AI/ML Model Training & Explainability (Day 1 - Night)

Engineer tabular + graph features: transaction velocity, fee anomaly ratio, peeling hop count, clustering coefficient, and night-time activity ratio. Train an Isolation Forest model for anomaly scoring and a Graph Neural Network / Gradient Boosting classifier for malicious typology prediction. Integrate TreeSHAP to compute top contributing factors.

Phase 4: Dashboard & Interactive Link Analysis (Day 2 - Morning)

Build the FastAPI endpoints (`/api/leads`, `/api/graph/{id}`, `/api/stats`). Connect the React frontend with Cytoscape.js. Enable clicking on any wallet node to view its associated IP origin, SHAP radar chart, and step-by-step transaction hops.

Phase 5: Packaging & Evaluation Dossier (Day 2 - Afternoon)

Package the entire solution in a single offline Docker container or standalone shell script (`run_offline.sh`). Test with sample data while physically disconnecting Wi-Fi/Ethernet to guarantee 100% offline compliance for NTRO judges.

7. Strategic Advice: What Will Make You Win SIH 2026

- **1. Strict Offline Guarantee:** Judges from NTRO are technical defense and intelligence experts. During the presentation, unplug the internet cable or turn off Wi-Fi before running the demo. A live web query to a third-party explorer will instantly disqualify a team; an offline system executing locally will earn top marks.
- **2. Real ML vs. Fake Rules:** Many competing teams will write 10 `if/else` statements and call it "AI". Clearly show your ML training pipeline, the loss curve, your GNN architecture, and the SHAP feature importance plot to prove genuine data science rigor.
- **3. Correlation Proof:** Clearly demonstrate how your system bridges the *Network Layer* and *Blockchain Layer*. Show a specific case: *"Here is an unconfirmed P2P broadcast from IP 185.x.x.x in Country X; our timing correlator matched it with TXID 9a2f... which subsequently triggered a 4-hop peeling chain."*
- **4. Intelligence Dossier Export:** Provide a "Download Law Enforcement Report" button that generates an actionable, timestamped investigative lead sheet with hash signatures.